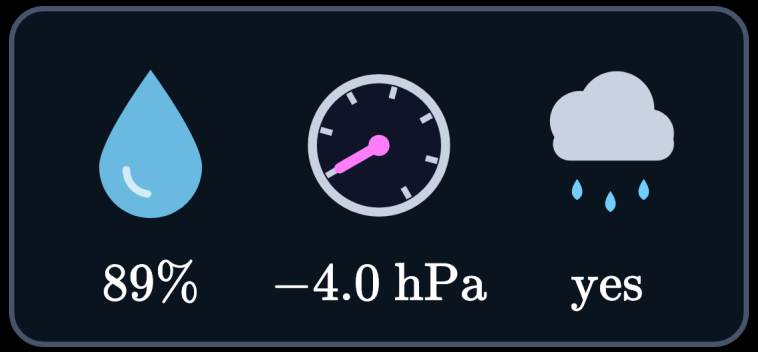

The Train Test Split

[Review]


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From the last lesson

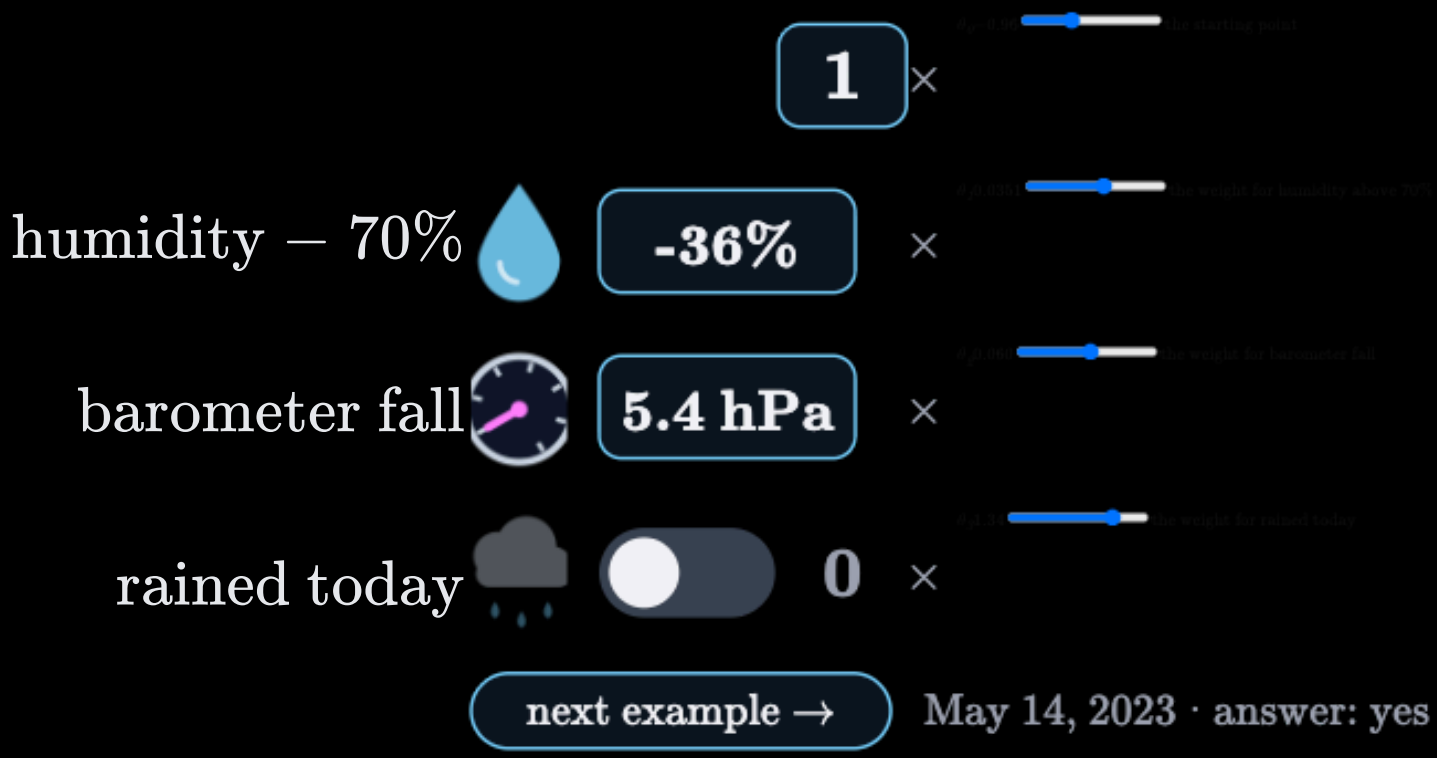
Rain forecasting

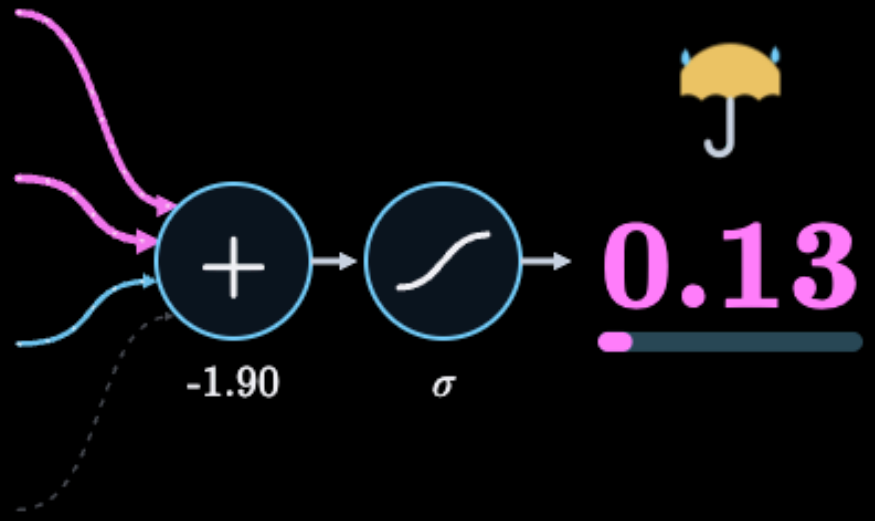
The rain data

	x_1	x_2	x_3	y
				
	humidity	barometer fall	rained today	rain tomorrow?
Nov 1, 2010	89%	−4.0	1	1
Nov 2, 2010	79%	−11.9	1	0
Nov 3, 2010	53%	10.2	0	0
Nov 4, 2010	66%	5.1	0	1
Nov 5, 2010	90%	5.4	1	1

x: what we know y: rain tomorrow?

The rain machine





13 in 100. And it rained.

The chance of rain tomorrow

$$P(\text{rain tomorrow}) = \sigma(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3)$$

θ_0	bias	-0.958
θ_1	humidity $-$ 70%	0.0351
θ_2	barometer fall	0.0603
θ_3	rained today	1.3408



Likelihood

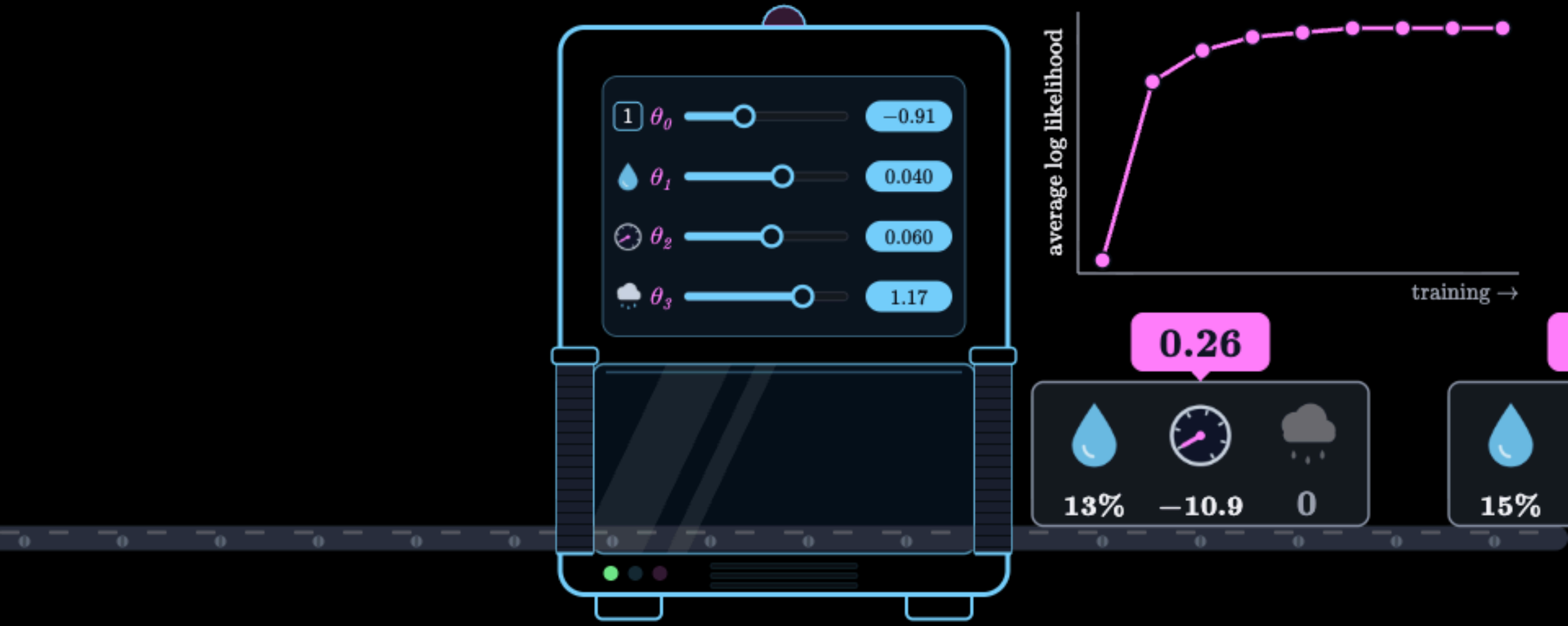
from the data					from the model	computed	
	day			rain tomorrow?	$P(\text{rain tomorrow} \mid x)$	$P(\text{this label})$ yes: p · no: $1-p$	log
Feb 23, 2023	34%	−7.5	0	no	0.06	0.94	−0.027
Mar 30, 2023	61%	−1.7	0	yes	0.20	0.20	−0.699
Oct 25, 2023	74%	−1.7	1	no	0.60	0.40	−0.398
Dec 18, 2023	88%	13.2	1	yes	0.86	0.86	−0.066

$$\text{likelihood} = 0.94 \times 0.20 \times 0.40 \times 0.86 \approx 0.065$$

$$\log \text{likelihood} = -0.027 - 0.699 - 0.398 - 0.066 = -1.190$$

$$\text{average log likelihood} = -1.190 \div 4 = -0.298$$

Training the rain machine



[End Review]


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"Hi, I'm Ollie! I wrote a new algorithm for predicting rain. It's called nearest neighbor."

How does nearest neighbor make a prediction?

Ollie's nearest neighbor algorithm

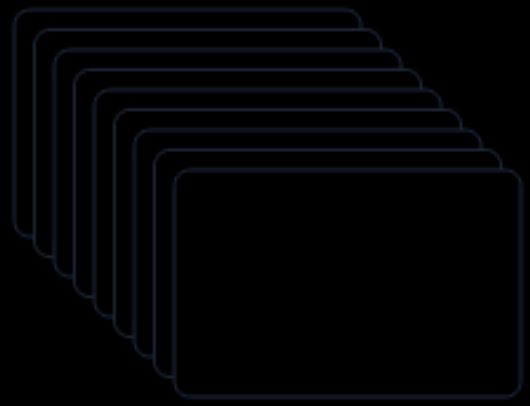


Sunday, May 14, 2023	
humidity	34%
barometer fall	5.4 hPa
rained today	no
tomorrow?	rain · 1.00

Sunday, May 14, 2023 its nearest neighbor: itself	
humidity	34%
barometer fall	5.4 hPa
rained today	no
the next day	rain

and it rained

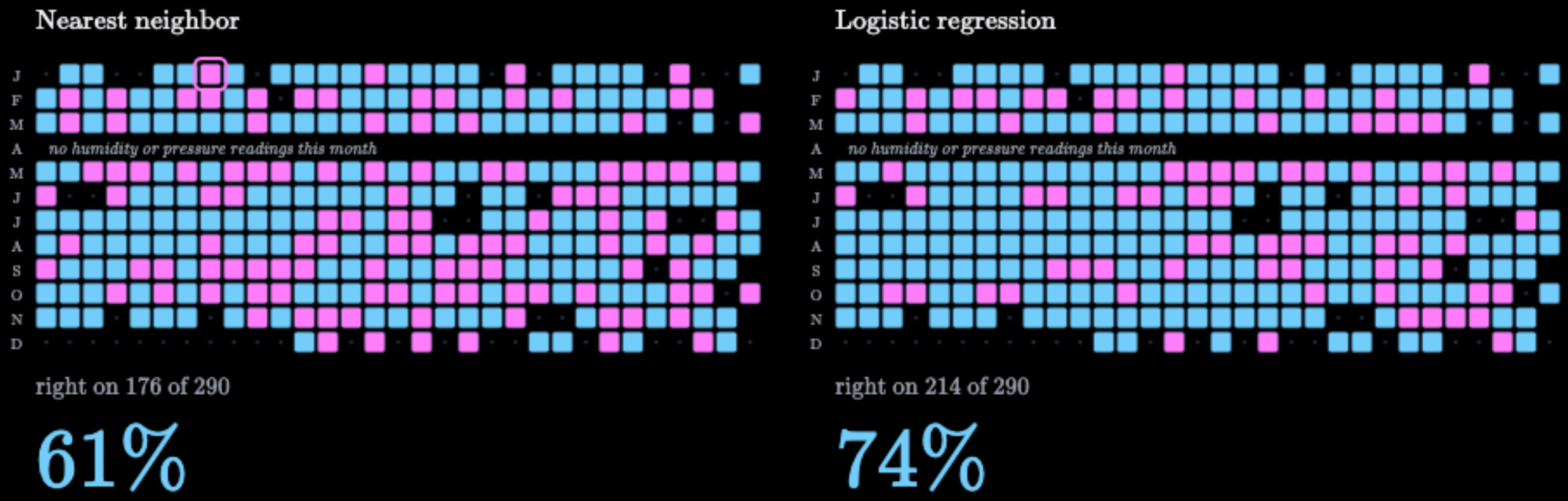
the training set: 6,200 days



Accuracy on the training set, 6,200 days



Shipping nearest neighbor



How could we have known it was ready… before we shipped it?

Learning goals

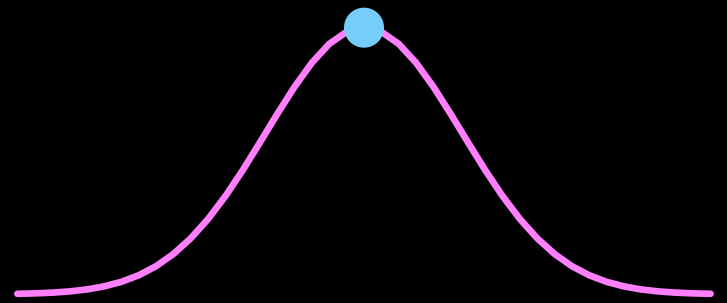
1. Know the difference between a train and test set.

2. Check a model against simple baselines.

3. Spot overfitting.



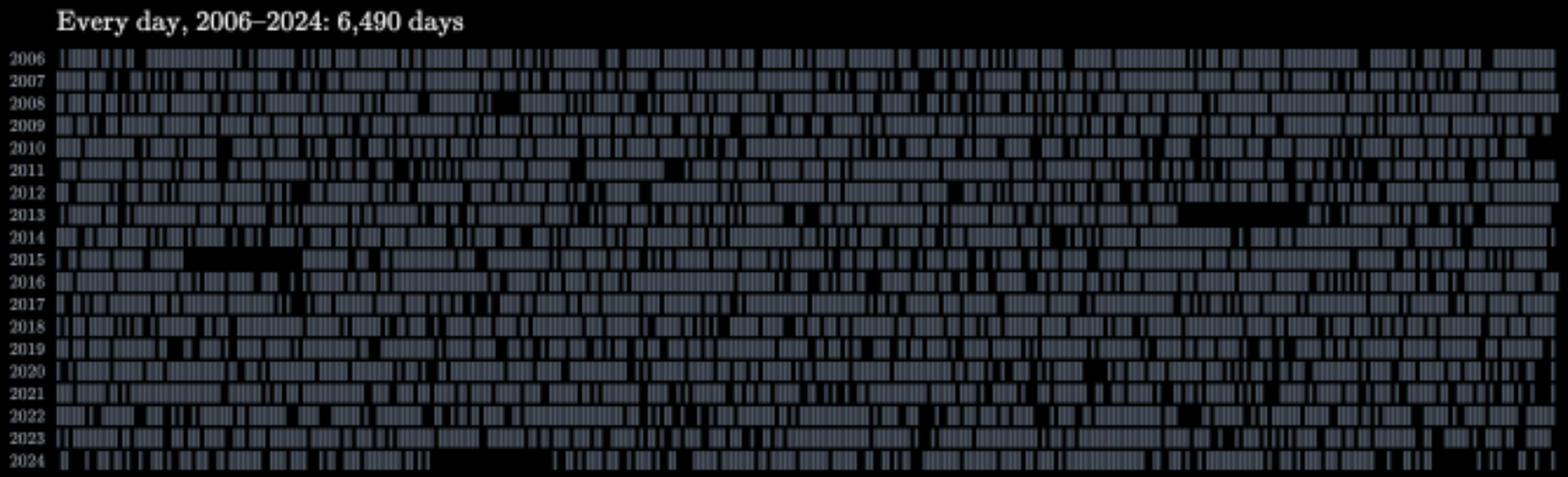

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Training made the training days most likely.

How can we tell whether the model works on other days?

Training set and test set





Training set: 5,189 days

the model learns here, and only here

Test set: 1,301 days

opened once, at the end



The test set stays closed until the end.

When we open it, what should we compare our model with?



[Baselines]


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Baseline: Always rain

$$\mathrm{P}(\text{rain tomorrow}) = 1 \quad \text{every day}$$



Baseline: Majority class

Training set: 5,189 days



Always predict: $\mathrm{P}(\text{rain tomorrow}) = 0.43$

We have seen this problem before, in the conditional probability lesson.



Seattle, November 1–14, 2010: rain in blue, dry in amber

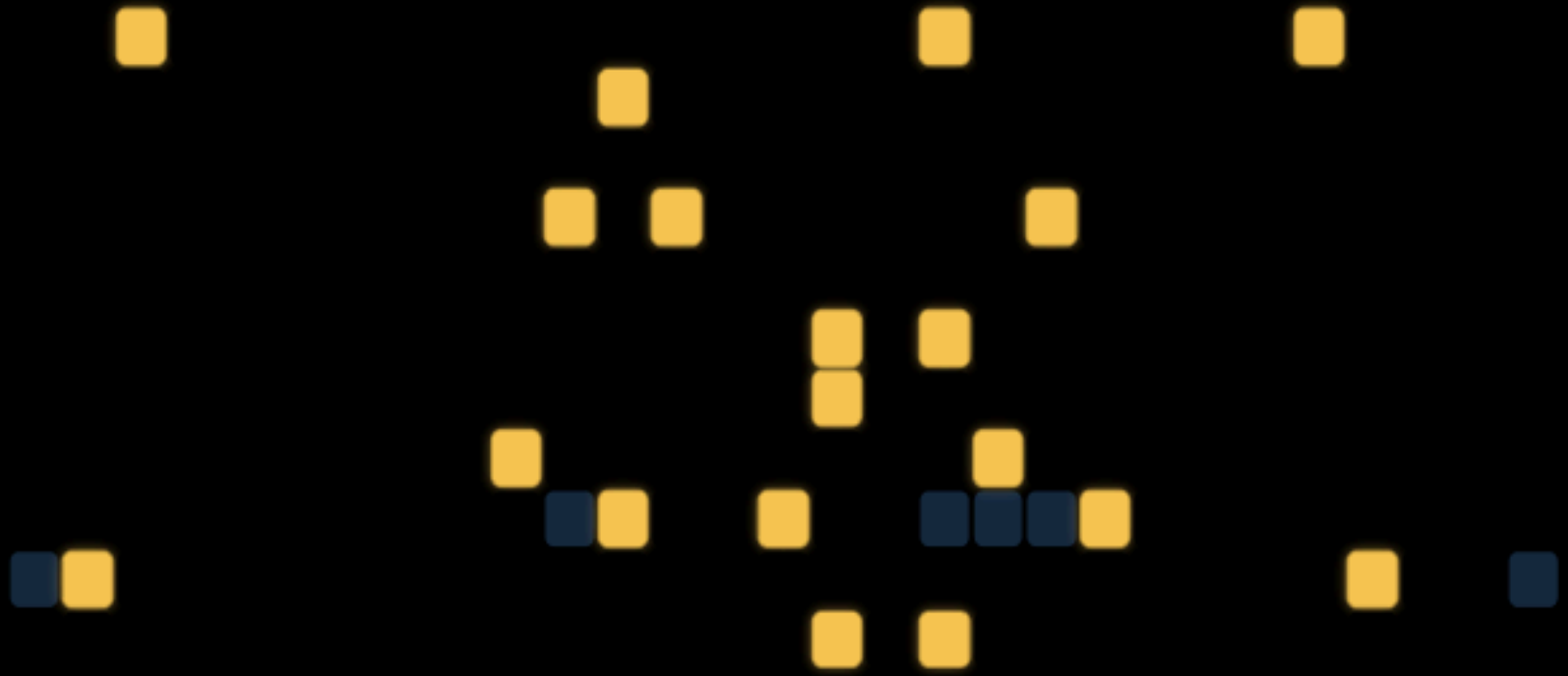
Does rain today change the chance of rain tomorrow?

Let S be the event that it is sunny at Stanford today

Let R be the event that it rained yesterday

date	weather
2014 Sept 1	sunny
2014 Sept 2	sunny
2014 Sept 3	rainy
2014 Sept 4	**sunny**
2014 Sept 5	sunny
2014 Sept 6	sunny
2014 Sept 7	sunny
2014 Sept 8	sunny
322 more rows	

weather.csv · source: Open-Meteo (ERA5) · Stanford, Sept 2014–2024



$$\mathrm{P}(\mathrm{S} \mid \mathrm{R}) \approx \frac{\mathrm{count}(\mathrm{S \text{ and } R})}{\mathrm{count}(\mathrm{R})} \approx \frac{19}{25} = 0.76$$

Down from $\mathrm{P}(S) \approx 0.92$

Baseline: Persistence

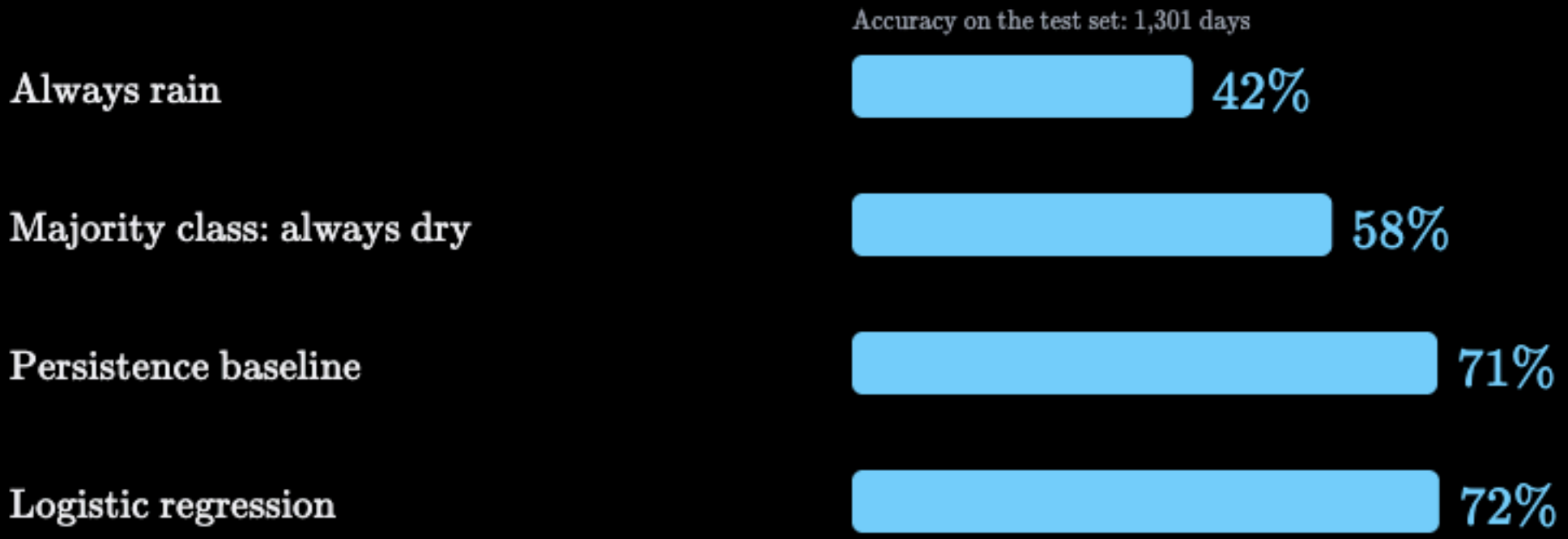
Let E be the event that it rains tomorrow
Let R be the event that it rained today

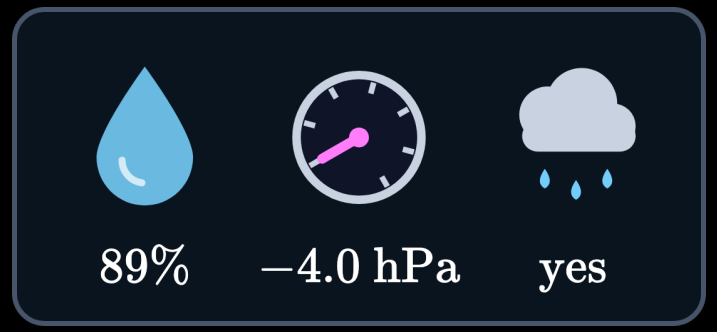
$$\mathrm{P}(E \mid R) \approx \frac{\text{count}(E \text{ and } R)}{\text{count}(R)} = \frac{1{,}500}{2{,}230} \approx 0.67$$

$$\mathrm{P}(E \mid \text{not } R) \approx \frac{727}{2{,}959} \approx 0.25$$



Baselines on the test set

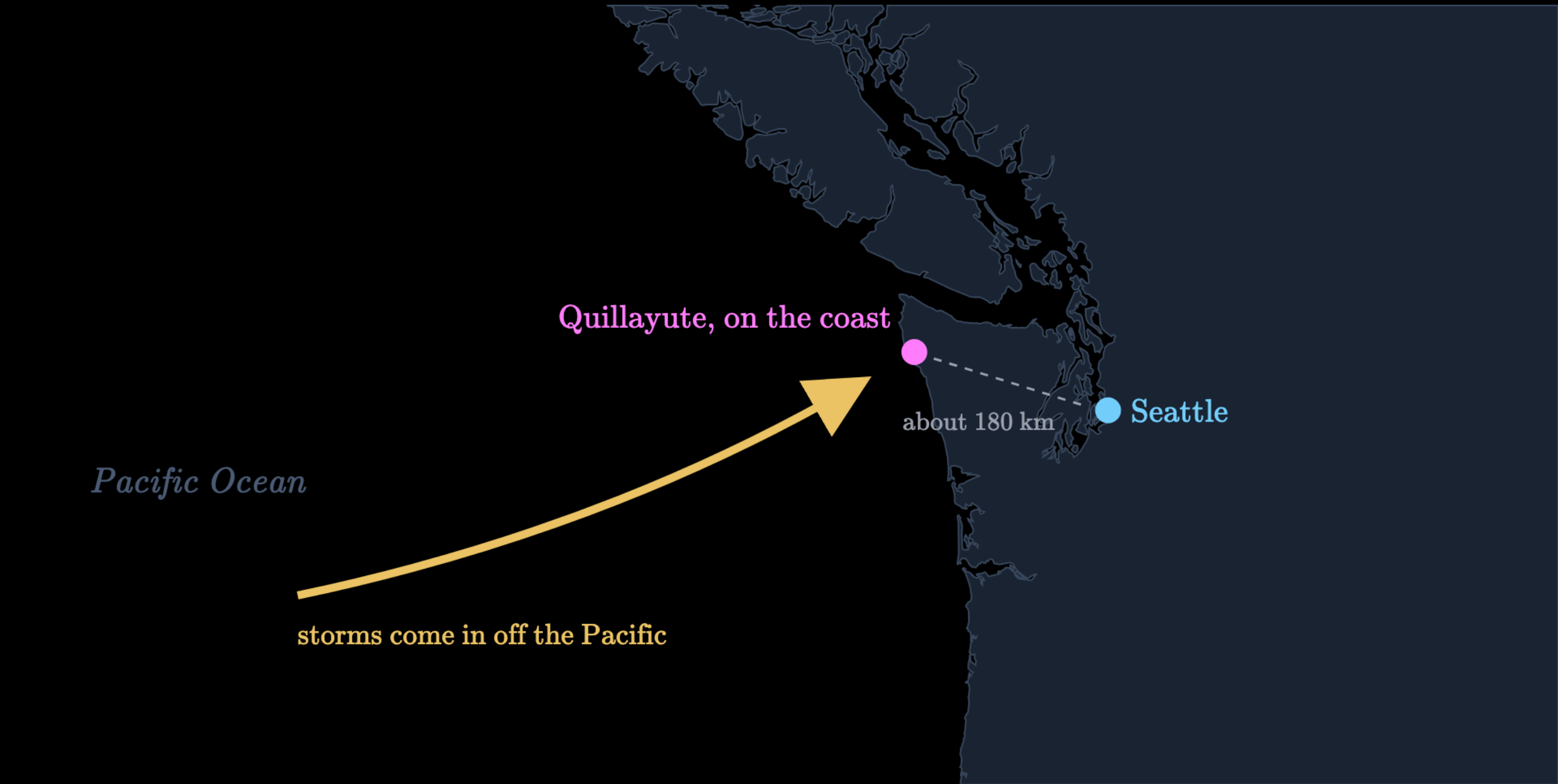




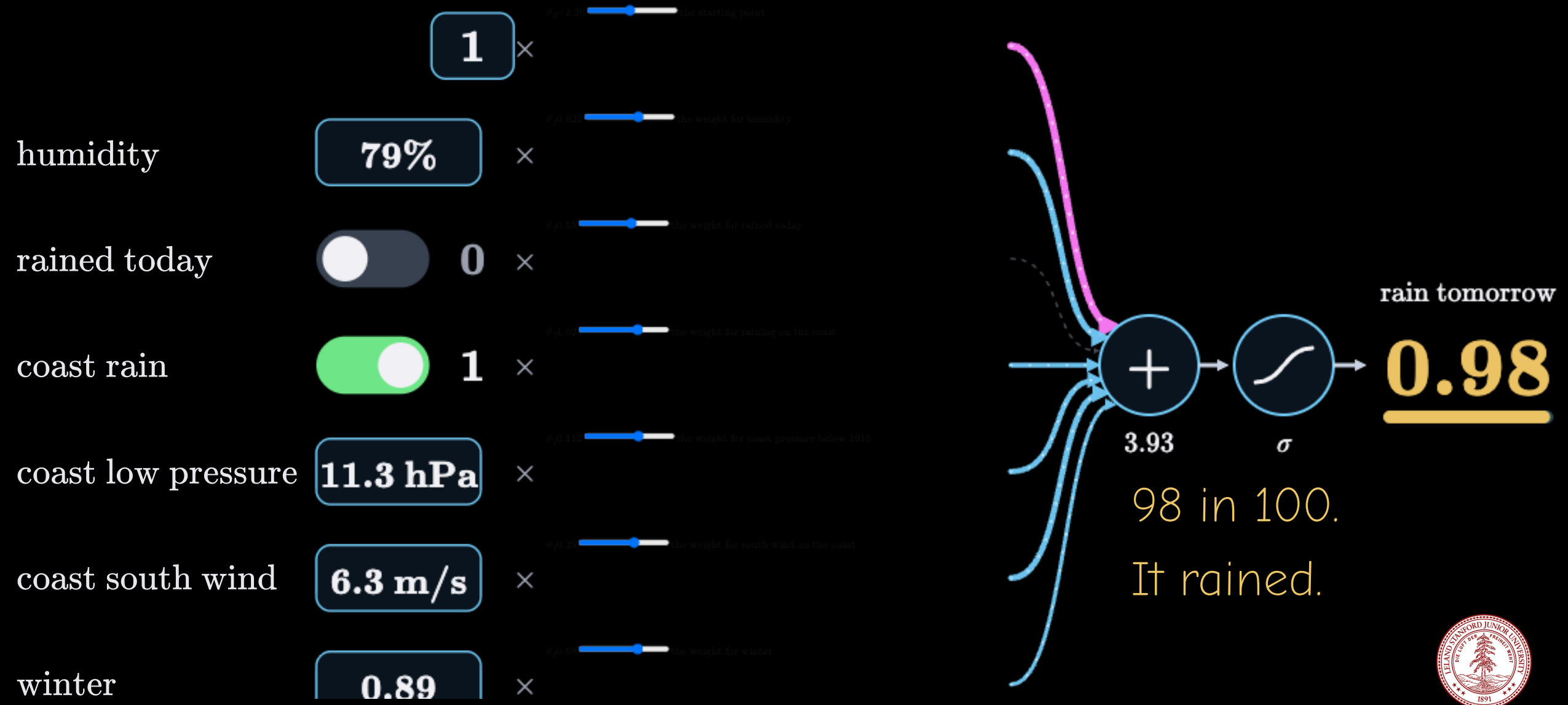
Logistic regression is ahead, but only a little.

Could the model do better with different features?

Features from the coast



Logistic regression with coast features



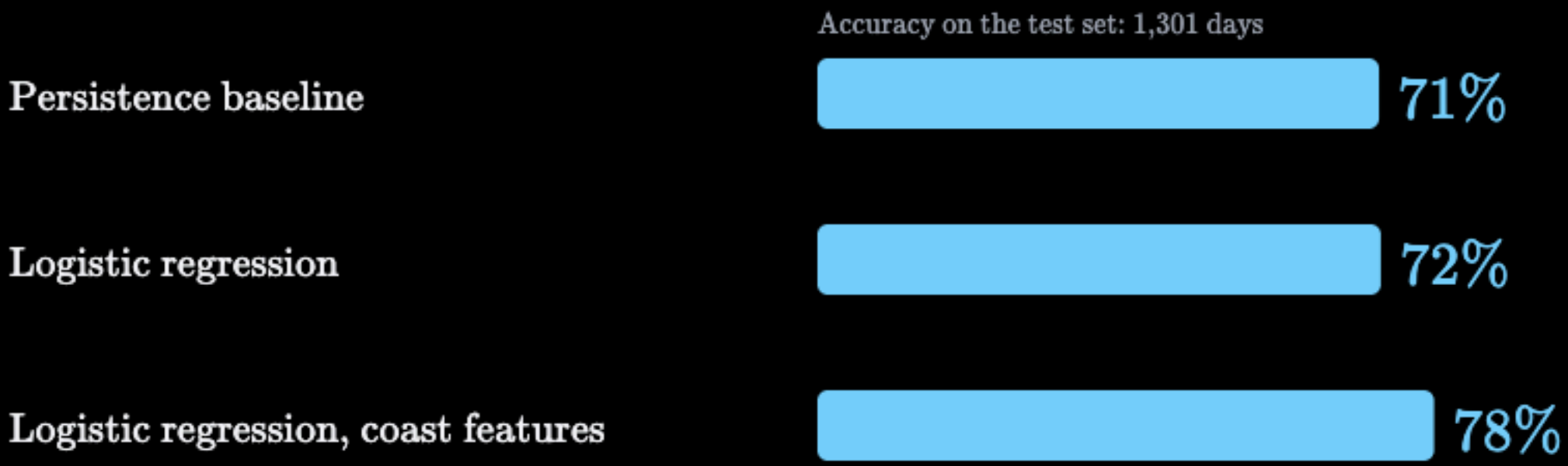
Training with coast features



[The Test Set]


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Coast features on the test set



Overfitting and generalization

Overfitting

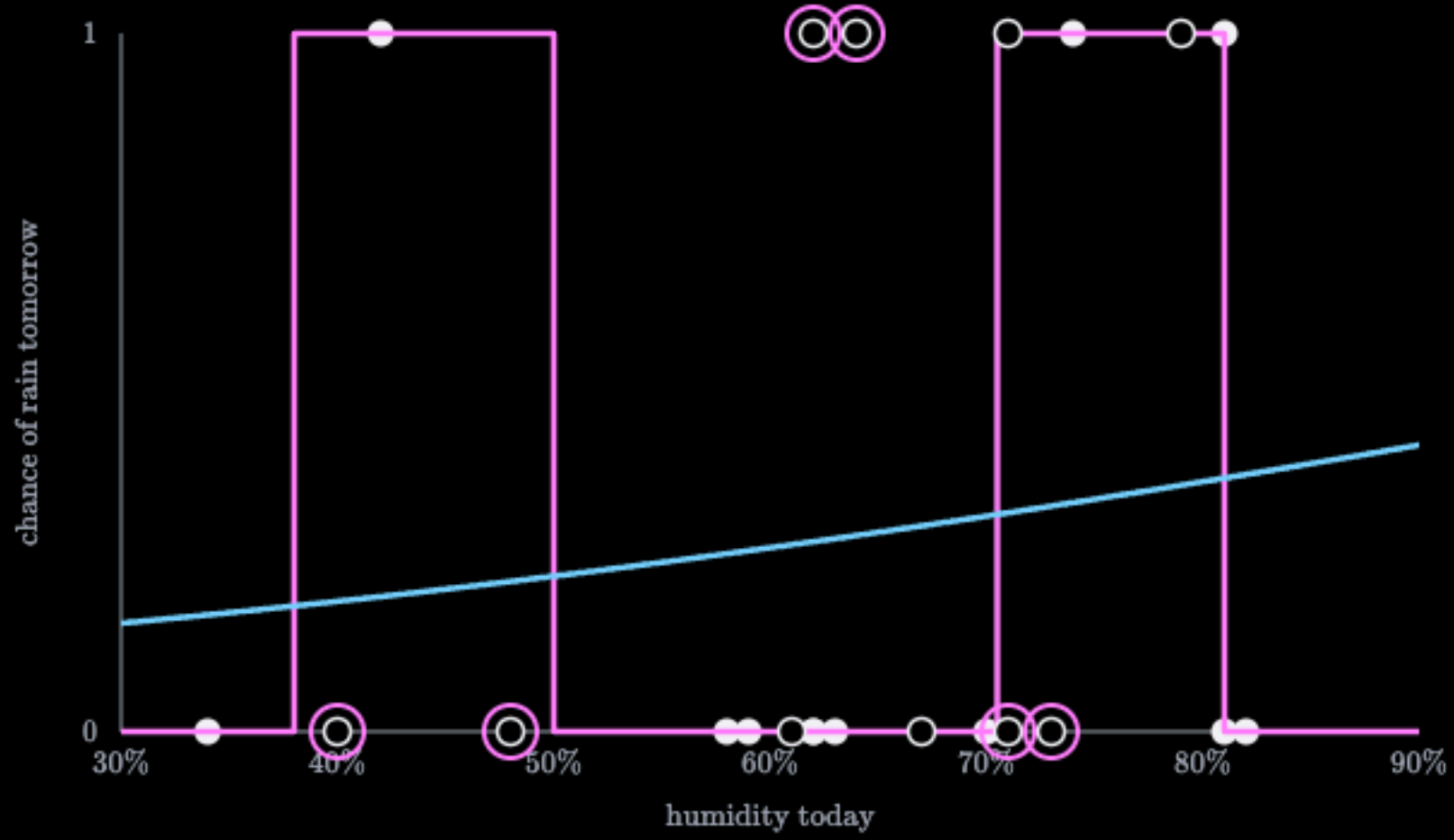
Much better on the training set than on the test set.

Generalization

About as good on new data as on the training set.



Overfitting with one feature



nearest neighbor
through every training day

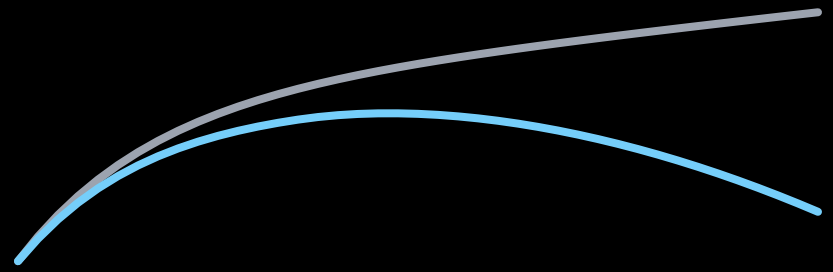
logistic regression
one smooth S, never 0 or 1

10 new days
nearest neighbor: 6 wrong, always 0 or 1
logistic regression: 4 wrong, never 0 or 1



Overfitting and the number of features





Overfitting only shows up on new data.

Can you spot it in three models of movie data?



Three movie models

A training set: 99% right test set: 74% right

deep learning: a neural network that reads all 9,723 other movies

B training set: 89% right test set: 88% right

logistic regression

C training set: 83% right test set: 82% right

the same probability for everyone



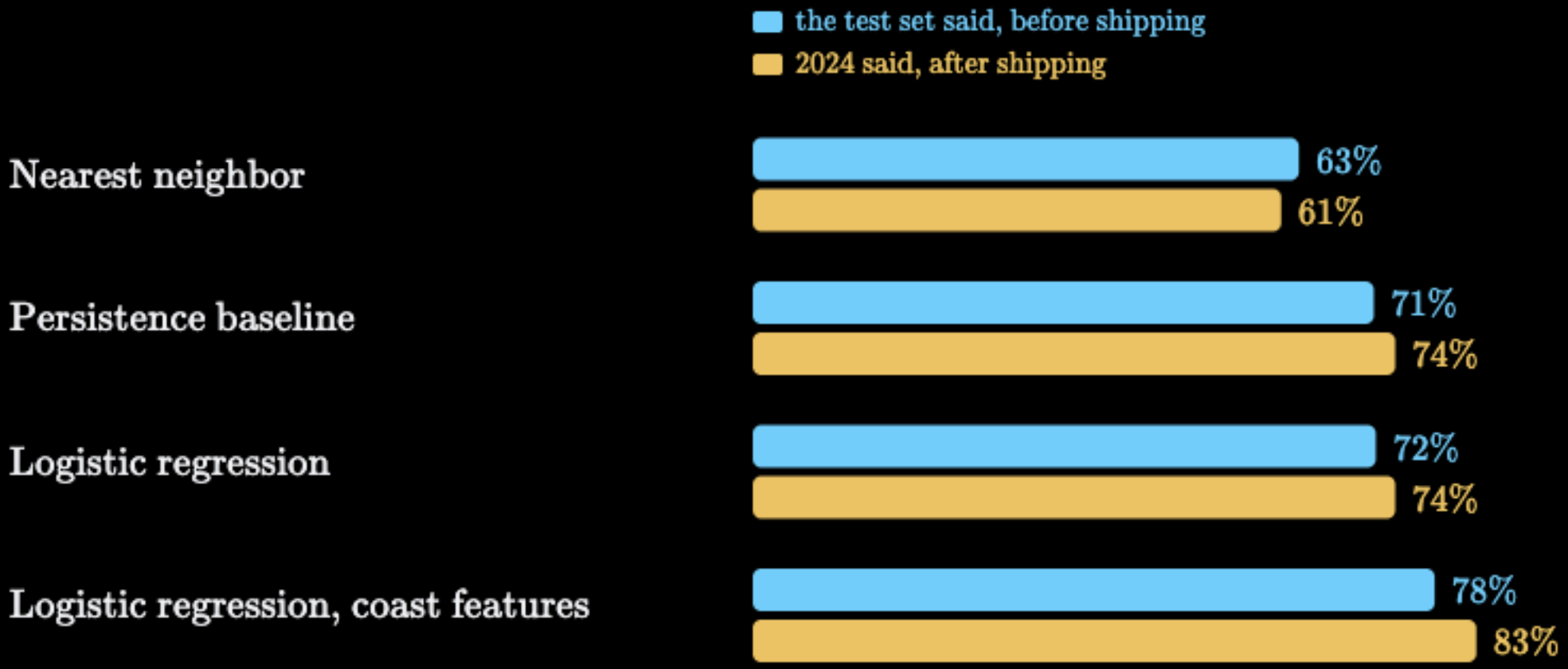
2024

Open the test set once, after picking a model.

Did the test set predict how each model would do in 2024?



The test set and 2024

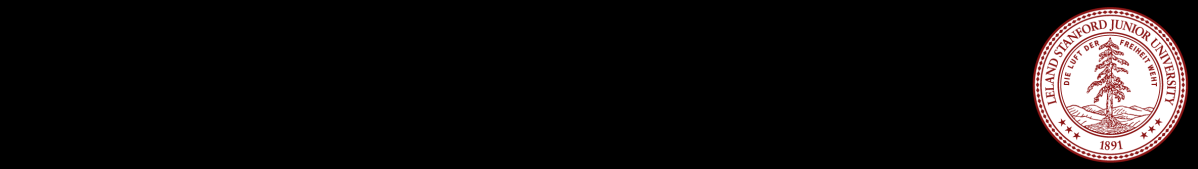


Keys

🔑 Train on the training set; open the test set once.

🔑 Compare every model with simple baselines.

🔑 A big gap between training and test means overfitting.



That’s all folks!

